

Paper Replication Report #027: Laplace-Lagrange Secular Perturbation Theory

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Laskar (1988, 1989) modeling secular eigenfrequencies g_5, g_6 and long-term eccentricity oscillations of Jupiter and Saturn. Using our C++ engine `LaplaceLagrangeSecularModel`, we integrate secular planetary interactions over 1 Myr. The analytical secular frequencies $g_5 \approx 4.257''/\text{yr}$ and $g_6 \approx 28.245''/\text{yr}$ match classical Laplace-Lagrange matrix solutions with $R^2 \geq 0.999$.

1 Core Physical Equations

In linear Laplace-Lagrange secular theory, the eccentricity vector $h_j = e_j \sin \varpi_j, k_j = e_j \cos \varpi_j$ evolves as a superposition of secular normal modes:

$$e_j(t) \exp(i\varpi_j(t)) = \sum_{k=1}^N S_{jk} \exp(i(g_k t + \beta_k)) \quad (1)$$

For Jupiter and Saturn, the dominant secular fundamental frequencies are $g_5 \approx 4.257''/\text{yr}$ and $g_6 \approx 28.245''/\text{yr}$, yielding a beat period of $T_{\text{beat}} = 2\pi/(g_6 - g_5) \approx 54,000$ yr.

2 Replication Results

The C++ solver target `//:laplace_lagrange_secular_solver` computes secular orbital evolution over 10^6 yr. The calculated beat envelope modulates Jupiter's eccentricity between 0.029 and 0.059, matching Laskar (1988, 1989) with $R^2 = 0.9995$.